Facility-Level Techno-Economic Potential of Electrified Industrial Process Heat in European Manufacturing [TODO: check title-length limit]

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Abstract

Industrial process heat dominates manufacturing emissions in the EU. Roughly 75% of energy used for industrial process heat is powered by fossil fuels. Yet, there are readily available solutions that can continuously generate process heat. Industrial heat pumps and thermal energy storage solutions paired with on-site solar PV can be a solution to bring power to heat generation quickly and cost competitively with natural gas. Using a dataset of 2,192 facilities of different manufacturing sectors from Climate TRACE, each facility is modelled with co-located, off-grid solar PV with industrial heat pumps and batteries installed in 2030. Under this modelled configuration, 59% of fossil-fuelled process heat can be decarbonized using solar PV within a 5 km buffer zone around each facility site at an average levelised cost of heat of €55/MWh. After expanding the buffer zone to 15 km, the share of possible decarbonized heat rises to 84% of process heat at €62/MWh against an EU average gas-heat cost of €93/MWh. Each facility is optimized to provide the highest availability and highest savings under its country's gas costs. These results show that Southern Europe can cost-competitively decarbonize over 95% of its heat demand. Central Europe (Rhine-Meuse Region), where most of the large, heat intensive sources are located, can decarbonize over 70% of its heat demand using on-site solar plus storage. [TODO: update numbers to final baseline run (85.0% / 59.6%?); keep ≤150 words]

Industrial process heat accounts for three quarters of industrial greenhouse-gas emissions in the European Union (EU)[1] and is an essential part of the climate change mitigation strategy[2]. The majority of industrial process heat demand is concentrated in the iron and steel, chemicals, and non-metallic minerals sectors, which also require the highest process temperatures, typically exceeding 1,000 °C[1].

Europe is facing challenges in energy security due to the reliance on international gas imports[3]. Diversifying energy sources and utilising renewable resources can not only reduce dependence on gas imports but also allow for quicker decarbonisation through electrification to meet EU climate goals.

Different technologies are already available to electrify process heat across different temperature heat bands such as heat pumps, industrial heat batteries, electric boilers, and electric arc furnaces. While industrial heat pumps are commercially available and highly efficient, they can only provide heat up to 200 °C[4]. Heat batteries (thermal storage) are a recent technology that are specified to provide heat up to 1500 °C, reaching temperatures needed for hard to abate industries such as

iron and steel and the chemical sectors[5]. Renewable generation using solar panels paired with heat pump and heat battery is a potential pathway to reduce carbon emissions and increase efficiency[6]. These technologies can generate replaceable process heat, heat that is generated with fossil fuel.

However, the progress to electrify process heat has been falling behind other sectors[7]. Critical barriers to electrifying industrial heat include high electricity grid prices relative to gas[7] and long lead times to connect to the electricity grid[8,9]. Off-grid renewable solutions can decrease reliance on imported gas[9] and avoid the multi-year grid interconnection timeline[10]. With enough land available, solar plus storage systems can provide heat without incurring the cost of expensive transmission infrastructure or long interconnection queue wait times[11]. [TODO: add from journal draft: industrial electricity price vs gas (€162 or €171/MWh – pick one) and queue lengths (3 years south; 5–10 years NL/DE); land/permitting trade-off.]

Prior work analyses industrial decarbonisation at a sector level or country level[3,12]. Facility level analysis has been conducted but not for co-located off-grid solar plus storage in the EU with country specific costs and gas benchmarks[8,13]. For on-site solar resources, the capacity factor and costs can differ greatly within a country or region. Facility-level bottom-up analysis can tackle those gaps[14] by using site-specific solar capacity factors, land availability, and country adjusted cost of infrastructure. [TODO: add: Gilbert et al. LCOH / grid-based conclusion[15]; Ziegler et al. cost vs reliability[16]; McMillan; Saad et al.; Markussen. Literature table → Supplementary Table 1.]

With facility-level resolution, this study analyses 2,192 EU facility with 5 and 15km land buffers to determine levelised cost of heat (LCOH) at each facility and quantifies the potential to decarbonise with the solar plus storage solution that is below the gas LCOH. [TODO: P6: rewrite as a "Here we show..." paragraph with one sentence per headline result.]

Results

Total heat served and LCOH

[TODO: rewrite heading as the finding] The aggregate replaceable heat demand served is analysed in two ways: maximising savings and maximising heat demand. To maximise savings, a facility should aim for availability up to the maximum savings points. The remaining can be met with grid electricity or gas. To maximise heat demand served, a facility could overbuild solar panels while still being under gas prices resulting in a lower marginal benefit but is still gas competitive. Table 1 summarises both framings at the 5 km and 15 km buffer zones.

At 5 km, 59% of replaceable heat is below country gas at €55/MWh, expanding to 15 km reaches 84% at €62/MWh. Stopping at max savings leaves more money on the table but only 53% and 74% of heat for 5 km and 15 km buffer zones respectively. The LCOH at 90% for solar plus storage is still almost €100 under the grid electricity LCOH (Fig. 2).

Country-level comparison

[TODO: rewrite heading as the finding] Germany, the Netherlands, and France have very large process-heat demand. With a ∼15 km land buffer, a large majority can be served below country gas. At 5 km, Germany and the Netherlands remain more land constrained. Southern Europe (Greece, Spain, Italy, Portugal) is the strongest fit. Those sites tend to be smaller, with better solar yield. ∼95% of demand served at LCOH roughly 70–75% of country gas cost (Fig. 3).

High temperature sectors, solar resource and land constraint

[TODO: rewrite heading as the finding; this subsection merges WISO26 §3.2, §3.4 and §3.5 – trim] Steam cracking and iron and steel sector have very large sites and have more heat intense processes (>500 °C) resulting in much higher heat loads. These sectors require more hybrid solutions: increased land availability, utilising grid electricity, co-firing with gas plants. By increasing buffer zone around each site, over 80% of heat can be served in both sectors under gas prices. Increasing buffer zones from 5 km to 15km, availability for steel increases from 40% to 80%, for steam, from 30% to 80%. Grid electricity or gas plants are still needed to provide for the last 10-30% of demand at 15 km buffer zones.

Where land is comparable, solar resource drives how much PV overbuild high-temperature sites need. Two high temperature steel plants, Linz in Austria and Taranto in Italy, have similar temperature needs (900 °C) and heat demand (5 TWh). Taranto has 20% solar capacity factor while Linz has 15%. Both have enough land available at 15 km to meet 90% of their heat demand, but Linz needs to overbuild 15x load while Taranto would only need to build 5.3x their load. Higher-irradiance sites can hit high-T reliability with less overbuild and much lower LCOH. Table 2 summarises the comparison.

For large sites with high temperature processes, there are concentrated areas where the land resources just cannot provide most of the energy at 5 km buffer zone. The major shortfall areas are in the Rhine-Meuse region where steel and steam cracking facilities are. These sites typically have ∼8–50 km^2/TWh land and ∼12–14% solar capacity factor. Most sites that miss 50% at 5 km recover at 15 km. Expanding to 15 km allows all sites, except for 3, to reach over 50% availability (Fig. 4).

Discussion

One of the major roadblocks to industrial electrification is the cost and delay to connecting to electricity grid infrastructure. However, with on-site renewable site, there is enough availability served such that industrial sites can use those resources. This study examines how much heat generation can be served on-site without connecting to grid infrastructure and how much that would cost. With even 5 km of buffer zone, 58.8% of replaceable heat demand in the EU can be met with solar + heat pump/heat battery at a cost below natural gas fuel costs with carbon tax. This solar + storage solution does not require the whole system to be decarbonized, nor a new

interconnection, for meaningful cost and emissions reductions. Facilities that cannot reach high availability on-site can still build to the maximum-savings point (53% of fleet heat at 5 km) and meet the remainder with gas or grid electricity. [TODO: add contrast with grid-based conclusion of Gilbert et al.[15]]

The interaction between solar radiance and land availability makes the cost considerations different for each site. The decision to maximise cost savings versus maximise heat demand served depends on how much solar panel overbuilding might be enough to justify the cost and benefit at each site. Where land is comparable, solar resource sets the overbuild. The Linz and Taranto steel plants have similar heat demand (∼5 TWh) and both can meet 90% availability at 15 km, but Linz (15% capacity factor) needs 15× load versus 5.3× at Taranto (21%). Those sites are the natural candidates for a hybrid remainder: gas, or grid electricity where industrial tariffs fall. In the Rhine-Meuse region, where high temperature and large industrial facilities are clustered. Land is the binding constraint. Regional regulation and electricity prices might help with decarbonisation by enabling cheaper electricity prices to meet part of the heat demand with grid electricity. Some countries in the EU have begun changing electricity prices for industrial customers. Germany has implemented state subsidised electricity prices that would help support electrification efforts[17]. Even with cost competitive grid electricity, the impacts on the distribution and transmission grid might prevent electrification from being approved or delay the electrification. With the cost of solar panels and storage falling rapidly, this could be a pathway to quicker and sooner decarbonisation.

[TODO: add: land and permitting as the policy lever (acceleration areas; cropland restrictions in France and Italy).]

Heat demand estimations are based on literature sector energy consumption factors and emissions factors, not based on individual facility's generation. These estimates are calculated based on EU published figures and averages by sector. They are validated by sector and country aggregated figures but may deviate from reference individual facilities significantly. With more precise data on combustion plants and fuel types of each facility, the modelled heat generation and LCOH can be better fitted per facility using the same process in this study.

This study does not account for the land acquisition process as there is not enough data to systematically incorporate into this study. Only remote sensing categorised data were used to screen available land. Each facility would need to evaluate the land acquisition process to accommodate the land needed for solar panels. Production data for glass and some other facilities are from 2014, and more recent facility-level emissions data, such as the EU Emissions Trading System and EEA industrial reporting, could be used in future work to update these estimates and cover facilities that currently have no heat estimate. [TODO: add remaining limitations: flat load; delivery medium not modelled; representative days (8,760-h check); single weather year; 2025 gas vs 2030 costs.]

On-site solar with heat pumps and heat batteries can serve most replaceable EU industrial process heat at costs below the 2025 gas-heat benchmark. Within a 5 km buffer zone, 58.8% of replaceable heat demand can be served at an average levelised cost of €55/MWh. That rate can be increased to 84.4% at €62/MWh with a 15 km buffer zone. Southern European facilities

tend to have stronger solar resource and less heat intense industrial facilities which allows them to serve almost their entire replaceable heat demand. The large, land-constrained, high-temperature sites are concentrated in the Rhine-Meuse region which will require hybrid solutions with either grid electricity or gas use to meet all its needs. Because this pathway is cost-competitive without requiring full decarbonisation of the heat demand or a new interconnection, it offers industrial facilities a deployable, near-term solution to reduce emissions and lower exposure to gas price volatility.

Methods

Facility dataset

We built a facility-level dataset of EU manufacturing facilities from the Climate TRACE emissions inventory[18]. For each facility it provides the location, subsector, annual production capacity and capacity factor, and annual greenhouse-gas emissions. The dataset covers cement, lime, iron and steel, aluminium and other non-ferrous metals, glass, chemicals, ethylene steam cracking, pulp and paper, food, beverages and tobacco, and textiles. There are 2,415 facilities in the EU in this dataset, of which 2,192 facilities have production or emissions data and therefore a replaceable heat estimate. The 223 facilities without an estimate are mainly glass (193 facilities) and other metals (27 facilities), for which neither source has production or emissions data.

Climate TRACE contains duplicate and placeholder records, so we cleaned the data before use. Duplicates were identified by identical facility names, coordinates, or capacity values, and by known ownership changes, and were reviewed against the sector totals in the top-down data described in Supplementary Note 1. In total, 78 records were removed across 67 groups of duplicate or double-listed records, for example cement plants that were also listed as lime. One removed record, a second listing of a glass site, was first merged into the kept record by adding its production so that the site's production is not lost. Identical capacity values across facilities are often a country-level default and not evidence of duplication, so these were not treated as duplicates on their own.

Annual production is taken from Climate TRACE, as capacity multiplied by capacity factor, for 540 facilities (73% of replaceable heat). Climate TRACE has no production for glass, and for some iron and steel and pulp and paper facilities it reports the same default value for every facility in a country. For these we use production from the Hotmaps industrial database[13] (2014 data), which applies to 172 facilities (148 glass, 16 iron and steel and 8 pulp and paper facilities; 5% of replaceable heat). Facilities are matched to Hotmaps records by location (within 5 km) and by name, and low-confidence matches are not used. Food and textile facilities (868 and 612 facilities respectively, 22% of replaceable heat) have no production data in either source, so their heat is estimated from emissions instead (see Replaceable process heat). Facility data therefore carry the vintage of their source, 2024 for Climate TRACE and 2014 for the Hotmaps substitutions. The Hotmaps substitutions are 2014 estimates, so glass and other sites that have opened, closed or changed capacity since are not captured. These substitutions affect 5% of replaceable heat.

Replaceable process heat

The replaceable process heat for each facility is the heat generated by fossil fuel that is used in the process of manufacturing a product (steel, food, textile, etc.). Depending on data availability, we estimate process heat from either total annual production or total annual emissions of each facility, as described above. Production is preferred where available because emissions are a less direct measure of process heat. Emissions include process emissions that do not come from fuel (for example, 60–65% of cement CO_2 is from calcination), and they depend on the fuel mix, process design and system boundaries of each site, such as on-site electricity generation. Manz et al. compared production-based and emission-based estimates of fuel demand for individual sites and found that production data are more accurate because process energy demand is more closely correlated with physical production than with emissions[19]. Emission-based estimates remain valid where production data are not available, which we use for food and textiles. Where production is available, a sector energy consumption (SEC)[13], the fuel input per tonne of product, converts production into fuel energy, and a process efficiency converts fuel energy into useful heat (Extended Data Table 1). Where only emissions are available, an emission factor[20] converts emissions into fuel energy, assuming an EU-average gas/oil share of 90%/10%[21]. Finally, applying the fossil-fuel share of each sector from Eurostat[22] to the useful heat gives the replaceable process heat. For facilities with production data, fuel energy and useful heat are

$$Q_{\mathrm{fuel}} = \frac{P \cdot \mathrm{SEC}}{1000}, \qquad Q_{\mathrm{useful}} = \eta\, Q_{\mathrm{fuel}}, \tag{1}$$

where P is annual production (t), SEC is the fuel input for process heat ($\mathrm{GJ\,t^{-1}}$), Q is in TJ and η is the useful thermal efficiency of the process. For facilities with only emissions data (food and textiles),

$$Q_{\mathrm{fuel}} = \frac{E_{\mathrm{CO_2e}}\, f_{\mathrm{comb}}}{\mathrm{EF}}, \qquad Q_{\mathrm{useful}} = \eta_{\mathrm{boiler}}\, Q_{\mathrm{fuel}}, \tag{2}$$

where $E_{\mathrm{CO_2e}}$ is annual emissions ($\mathrm{tCO_2e}$), f_{comb} is the share of emissions from fuel combustion (1.0 for food and textiles, which have no process CO_2) and EF is the fuel emission factor ($\mathrm{tCO_2\,TJ^{-1}}$). EF is the energy-weighted mix of the emission factors of natural gas ($56.1\,\mathrm{tCO_2\,TJ^{-1}}$) and residual fuel oil ($77.4\,\mathrm{tCO_2\,TJ^{-1}}$)[20], at a gas/oil energy share of 90%/10%, which is the fossil fuel mix of EU food and textile industries in Eurostat energy statistics[21]. This gives $0.9 \times 56.1 + 0.1 \times 77.4 = 58.2\,\mathrm{tCO_2\,TJ^{-1}}$. Replaceable heat is then $Q_{\mathrm{rep}} = s_{\mathrm{fossil}}\, Q_{\mathrm{useful}}$, with s_{fossil} the fossil-fuel share, the fossil fuel use divided by the sum of fossil fuel and biomass use for the industry in that country, from Eurostat disaggregated final energy consumption in industry (nrg_d_indq_n, 2024)[22]. Fossil fuels include coal, oil, natural gas and non-renewable waste, and biomass includes biofuels and renewable waste. Purchased electricity and derived heat (heat bought from outside the site, such as district heating) are excluded from the ratio, which assumes that purchased heat has the same fossil share as on-site fuels. The heat demand itself is not reduced. The EU-27 value, calculated from the summed fuel use of all 27 member states, is used where a country value is unavailable. SEC and efficiency

values by sector are in Extended Data Table 1, and emission-route parameters in Extended Data Table 2.

Temperature bands

Each facility's useful heat is also split into five temperature bands (<100, 100–200, 200–500, 500–1000 and >1000 °C), which determine the technology that can serve it. The band shares come from the Hotmaps industry process benchmarks[13], which give the temperature distribution of fuel use for each sector and process, and are applied to the useful heat of each facility. For facilities with production data, the shares come from the sector's process route (for example sinter and rolling for blast-furnace steel). For food, where only emissions are available, the facility is matched by name to a Hotmaps process (sugar, dairy, brewing, meat, bakery or starch) and defaults to dairy when no match is found. Textiles are not split by process. Their temperature split is taken from the EU-27 textile and leather useful heat in the IndustryHeat-EU dataset[2], which has 32% of heat below 100 °C, 68% as steam, and no demand above 400 °C. We assign the steam to the 100–200 °C band, so all textile heat is served by heat pumps.

We performed additional validation against sector and country aggregates of useful heat from JRC-IDEES[23] to confirm that the heat estimates are reasonable (Supplementary Note 1).

Solar resource

Hourly solar generation for each facility is taken from the Photovoltaic Geographical Information System (PVGIS)[24] using ERA5 reanalysis[25] for the calendar year 2023. One time series is requested per facility at its latitude and longitude, for a 1 kW_p system, so that the output is per unit of DC nameplate capacity. Panels are fixed-mounted and south facing, with a tilt equal to the absolute latitude of the site, which approximates the annual-energy-maximising angle without optimising each site. The PVGIS series are pulled with no system losses, and a 14% system loss, the default of PVGIS and of the NREL PVWatts model[26], is then applied in the model by reducing the hourly output by 14%. The loss covers inverter, wiring, soiling and other losses, and PVGIS accounts for temperature and irradiance effects separately. The capacity factor reported for each site is the annual sum of the hourly output divided by 8,760 h. Annual yields before losses range from about 950 kWh kW_p^{-1} in northern Finland to 1,950 kWh kW_p^{-1} in southern Portugal. Because the model sizes storage to cover day-to-day operation rather than seasonal storage, a single weather year is used. The 365 daily solar profiles of 2023 are clustered into 16 representative days (k-means, using each cluster's medoid day, weighted by cluster size) so that sunny, overcast and intermediate days across the seasons are all represented (see Optimization model). Year-to-year variation in solar resource is therefore not captured.

Land availability

Land availability sets the maximum solar capacity that can be built at each facility. We define the land area available to each facility as the eligible land within a circular buffer centred on the facility coordinates, using a 5 km radius for the main analysis and 15 km as a sensitivity. We calculated this using Google Earth Engine at 30 m resolution. Land cover is taken from the ESA WorldCover v200 map for 2021[27], and the classes counted as available are bare or sparse vegetation, cropland, grassland and shrubland. Built-up areas, tree cover, water, wetland and snow and ice are excluded. Grassland and shrubland are included because many mills, particularly Nordic pulp and paper sites, sit in forest and grassland mosaics where bare land and cropland alone report almost no usable area.

Two further screens remove land that is unsuitable for ground-mounted solar. Pixels with a slope above 5° are excluded, using SRTM elevation data and Copernicus GLO-30 above 60°N, where SRTM has no coverage. Protected areas in the World Database on Protected Areas (WDPA)[28] with a status of designated, inscribed or established are also excluded, together with a 100 m buffer around them. A power density of $50\,\mathrm{MW_{dc}\,km^{-2}}$ is applied to the remaining area to set an upper limit on installed solar capacity at each site. Land enters the optimisation only as a hard capacity constraint, and land cost is assumed to be included in the operating costs of the BloombergNEF technology cost data[29]. Facility buffers are assessed independently, so overlapping buffers of neighbouring facilities are not allocated between them and land could be counted by more than one site.

Technology costs and financing

All costs are expressed in constant (real) 2025 US dollars for a 2030 construction start, and technologies are assumed to have a 25-year life. Capital expenditure (CAPEX), fixed operation and maintenance (FOM) and variable operation and maintenance (VOM) of solar panels and utility-scale Li-ion batteries are taken from BloombergNEF (BNEF)[29] for 2030 by country, reported in real 2025 dollars. BNEF reports direct data for only a few countries: five of the countries in our dataset for solar PV (Germany, Spain, France, Italy and the UK) and two for batteries (Germany and the UK). All other countries are assigned the mean cost of peer countries in the same World Bank region and income group, falling back to the income group and then to the global mean. As a result, European battery costs outside Germany and the UK are the mean of the German and UK values. Battery VOM is charged per MWh discharged, and PV VOM is zero in the BNEF data. Land cost is not modelled separately and is assumed to be included in the BNEF operating costs.

Industrial heat pump costs by temperature band[30] and industrial heat battery (IHB) costs[31] are in 2021 and 2023 US dollars respectively. Both are converted to real 2025 dollars using US CPI, which applies factors of 1.19 and 1.06. No cost decline or increase beyond general inflation is assumed between the source year and 2030, so these technologies are priced at their published cost level, unlike the BNEF solar and battery costs, which are projected to 2030. This is a conservative choice, because costs of heat pumps and heat batteries are more likely to fall than rise by 2030 as deployment grows. Heat pump CAPEX is $\$832\,\mathrm{kW_{th}^{-1}}$ below 100 °C and $\$1{,}034\,\mathrm{kW_{th}^{-1}}$ at 100–200 °C, with FOM of \$14.3 and $\$21.4\,\mathrm{kW_{th}^{-1}\,yr^{-1}}$ (2025 USD), and coefficients of performance (COP)

of 3.7 and 2.2 respectively[30]. IHB CAPEX is $\$105.7\,\mathrm{kW}^{-1}$ for charging, $\$317\,\mathrm{kW}^{-1}$ for delivery and $\$5.3\,\mathrm{kWh}^{-1}$ for storage, with no FOM and a round-trip efficiency of 95%[31]. The Li-ion round-trip efficiency is set as an assumption at 90%, slightly above the current BNEF survey median of 87.5%, to reflect expected improvement by 2030. Degradation, replacement costs, grid purchases and curtailment costs are not included, and the system is assumed to operate for its full life at the initial performance. The key parameters are summarised in Extended Data Table 3.

Capital costs are annualised using the capital recovery factor (CRF) for each country, calculated over the 25-year life from the real weighted average cost of capital (WACC) from IRENA[32], consistent with the real-dollar costs. The solar PV WACC is used for all technologies. The UK uses its own BNEF costs and WACC. Costs are calculated in US dollars and the results are converted to euros at a fixed rate of 1.08 USD per EUR.

Total annualised system cost, C^{ann}, sums four technology terms (solar, Li-ion battery, industrial heat battery, and heat pump), each annualised with the CRF above. The Li-ion battery, industrial heat battery, and heat pump terms follow the same functional form using their respective CAPEX and FOM values described above. Each follows the same general form, illustrated here for solar:

$$C^{\mathrm{ann}}_{\mathrm{solar}} = \left(\mathrm{CAPEX}_{\mathrm{PV}} \cdot 1000 \cdot \mathrm{CRF}_{r_c} + \mathrm{FOM}_{\mathrm{PV}}\right) S^*, \qquad \mathrm{LCOH}_{\mathrm{solar}} = \frac{C^{\mathrm{ann}}_{\mathrm{solar}}}{Q_{\mathrm{served}}}, \tag{3}$$

where r_c is country WACC, S^*, E^*, P^* are optimised capacities, Q^*_b is HP thermal capacity sized to served heat in band b, and $Q_{\mathrm{served}} = Q_{\mathrm{demand}} - Q_{\mathrm{unserved}}$. Unserved energy is not costed (VoLL = 0).

Gas heat benchmark

Each facility's LCOH is compared against the cost of continuing to produce the same heat from natural gas, which we call the gas heat benchmark. The benchmark sets the highest availability of electrified heat that can be supplied below the gas heat cost, and each facility is capped at the gas heat cost of its own country. We assume the existing gas boiler is already installed and paid for, so the gas heat cost includes only fuel and carbon costs, and excludes boiler capital and maintenance costs. This favours gas, because the gas cost does not include the capital cost of the boiler.

Gas prices by country are from Eurostat gas prices for non-household consumers (nrg_pc_203)[33], taken for the large industrial consumption band of 100,000–999,999 GJ per year, excluding VAT and other recoverable taxes, and averaged over the two half-years of 2025. Cyprus, Luxembourg and Malta have no data for this band and use the EU-27 average. The price is divided by a gas boiler efficiency of 0.85[34] to give the cost per unit of useful heat, and a carbon cost is added for the emissions of burning the gas:

$$c^{\mathrm{gas}}_{\mathrm{heat},c} = \frac{p^{\mathrm{gas}}_c}{\eta_{\mathrm{boiler}}} + \frac{p_{\mathrm{EUA}} \times EF_{\mathrm{NG}}}{\eta_{\mathrm{boiler}}} \tag{4}$$

where p^{gas}_c is the gas price in country c (EUR/$\mathrm{MWh}_{\mathrm{fuel}}$), $\eta_{\mathrm{boiler}} = 0.85$ is the boiler efficiency,

$p_{\mathrm{EUA}} = 73.43\,\mathrm{EUR/tCO_2}$ is the 2025 EU Emissions Trading System allowance price[35], and $EF_{\mathrm{NG}} = 0.202\,\mathrm{tCO_2/MWh_{fuel}}$ is the CO_2 emission factor of natural gas ($56.1\,\mathrm{tCO_2\,TJ^{-1}}$ [20]). The carbon term adds a uniform $€17.5\,\mathrm{MWh_{th}^{-1}}$ in every country, so differences in the benchmark between countries come from gas prices. The resulting benchmark ranges from €65 to $€126\,\mathrm{MWh_{th}^{-1}}$ across the 27 countries, with a heat-weighted mean of $€80\,\mathrm{MWh_{th}^{-1}}$. Eurostat does not report UK gas prices, so the UK uses the Irish price. The benchmark is converted to US dollars at 1.08 USD per EUR for comparison with the LCOH. Gas prices are for 2025 and are not projected to 2030, which matches the real 2025 dollar basis of the technology costs, but ignores any change in gas or carbon prices by 2030.

Optimization model

For each facility, we size the co-located solar, storage and heat conversion equipment that delivers its process heat at the lowest levelised cost of heat (LCOH). The problem is a linear program solved with HiGHS, adapted from the framework of ref. 8 and illustrated in Fig. 1. It is solved separately for each facility over the 16 representative days of 2023 PVGIS hourly solar profiles (see Representative days), and each day is solved with a cyclic state of charge, so storage operates within a day only.

The heat demand of a facility is a flat load split across its temperature bands (see Temperature bands). Shared solar capacity supplies two routes. Bands below 200 °C are served by heat pumps, which can use solar electricity directly or through a Li-ion battery. Bands at or above 200 °C are served by an industrial heat battery (IHB), which is charged resistively from solar and discharges heat directly to the process. The model chooses the solar capacity, battery and heat battery capacity and heat pump capacity that minimise cost, subject to meeting a required availability, the share of annual heat demand that is served, and to the solar capacity the land allows. Symbols are defined in Supplementary Table 1.

The optimisation has two phases. In the first phase, the model minimises the LCOH while serving at least a required availability AF of annual heat demand. If land limits solar capacity so that this availability cannot be reached, the first phase is infeasible and the second phase is used instead: it first finds the maximum heat that can be served, and then minimises cost at that level of service.

Phase 1: Minimum cost at a required availability

Objective

$$\begin{aligned}\min\ \mathrm{LCOH} = \frac{1}{E_{\mathrm{load}}}\Big[\, & c_{\mathrm{PV}}P_{\mathrm{PV}} + c_{\mathrm{Li},E}E_{\mathrm{Li}} + c_{\mathrm{Li},P}P_{\mathrm{Li}}^{\mathrm{dis}} \\ & + c_{\mathrm{IHB,in}}P_{\mathrm{IHB}}^{\mathrm{ch}} + c_{\mathrm{IHB,out}}P_{\mathrm{IHB}}^{\mathrm{dis}} + c_{\mathrm{IHB},E}E_{\mathrm{IHB}} + \sum_{b\in\mathcal{B}^{\mathrm{HP}}} c_b^{\mathrm{HP}}P_b^{\mathrm{HP}}\Big]\end{aligned} \tag{5}$$

Each cost term is the annualised capital cost of the technology (CAPEX multiplied by the capital recovery factor) plus its operation and maintenance (O&M) costs. Solar includes fixed O&M, and variable O&M is zero in the BNEF data. The Li-ion battery includes fixed O&M on its discharge power and variable O&M on the energy discharged. The heat pump includes fixed O&M, and the heat battery has no O&M cost.

$$E_{\text{load}} = \sum_{b} L_b \sum_{k} w_k H \tag{6}$$

Constraints

Shared solar limit:

$$\sum_{b \in \mathcal{B}^{\text{HP}}} x^{\text{dir}}_{b,k,t} + p^{\text{ch}}_{\text{Li},k,t} + p^{\text{ch}}_{\text{IHB},k,t} \leq P_{\text{PV}}\, g_{k,t} \tag{7}$$

Storage state of charge:

$$s_{\text{Li},k,t} = s_{\text{Li},k,t-1} + \eta_{\text{Li}}\, p^{\text{ch}}_{\text{Li},k,t} - \frac{p^{\text{dis}}_{\text{Li},k,t}}{\eta_{\text{Li}}} \tag{8}$$

$$s_{\text{IHB},k,t} = s_{\text{IHB},k,t-1} + \eta_{\text{IHB}}\, p^{\text{ch}}_{\text{IHB},k,t} - \frac{p^{\text{dis}}_{\text{IHB},k,t}}{\eta_{\text{IHB}}} \tag{9}$$

$$\eta_{\text{Li}} = \sqrt{\text{RTE}_{\text{Li}}}, \qquad \eta_{\text{IHB}} = \sqrt{\text{RTE}_{\text{IHB}}} \tag{10}$$

Storage rate limit for each storage $j \in \{\text{Li}, \text{IHB}\}$:

$$s_{j,k,t} \leq E_j, \qquad p^{\text{ch}}_{j,k,t} \leq P^{\text{ch}}_j, \qquad p^{\text{dis}}_{j,k,t} \leq P^{\text{dis}}_j \tag{11}$$

Maximum storage duration:

$$E_j \leq h_{\max}\, P^{\text{dis}}_j \tag{12}$$

Li-ion discharge to HP bands:

$$p^{\text{dis}}_{\text{Li},k,t} = \sum_{b \in \mathcal{B}^{\text{HP}}} x^{\text{batt}}_{b,k,t} \tag{13}$$

HP heat balance and capacity:

$$\text{COP}_b \left(x^{\text{dir}}_{b,k,t} + x^{\text{batt}}_{b,k,t} \right) + u_{b,k,t} = L_b\, \sigma_t \tag{14}$$

$$\text{COP}_b \left(x^{\text{dir}}_{b,k,t} + x^{\text{batt}}_{b,k,t} \right) \leq P^{\text{HP}}_b \tag{15}$$

IHB discharge across temperature heat bands: the heat battery is one shared store, and its discharge is allocated across the bands it serves:

$$p^{\mathrm{dis}}_{\mathrm{IHB},k,t} = \sum_{b \in \mathcal{B}^{\mathrm{IHB}}} y_{b,k,t}, \qquad y_{b,k,t} + u_{b,k,t} = L_b \, \sigma_t \tag{16}$$

Availability:

$$\sum_k w_k \sum_t \sum_b u_{b,k,t} \leq (1 - AF)\, E_{\mathrm{load}} \tag{17}$$

Bounds:

$$0 \leq P_{\mathrm{PV}} \leq P^{\max}_{\mathrm{PV}} \tag{18}$$

$$P^{\mathrm{ch}}_j,\ P^{\mathrm{dis}}_j,\ E_j,\ P^{\mathrm{HP}}_b,\ x^{\mathrm{dir}}_{b,k,t},\ x^{\mathrm{batt}}_{b,k,t},\ y_{b,k,t},\ p^{\mathrm{ch}}_{j,k,t},\ p^{\mathrm{dis}}_{j,k,t},\ s_{j,k,t},\ u_{b,k,t} \geq 0 \tag{19}$$

Phase 2: Maximum service, then minimum cost When the availability floor in Phase 1 cannot be met, the model first minimises the unserved heat U:

$$\min\ U = \sum_k w_k \sum_t \sum_b u_{b,k,t} \tag{20}$$

It then minimises cost with the unserved heat held at its optimal value U^*, so that the system is the cheapest that achieves the maximum possible service. Unserved heat is not assigned a cost in the objective:

$$\min\ \mathrm{LCOH} \quad \text{s.t. constraints (7)–(16), (18)–(19), and} \quad \sum_k w_k \sum_t \sum_b u_{b,k,t} \leq U^* \tag{21}$$

Nomenclature Symbols used in the optimization model are defined in Supplementary Table 1.

Representative days

The optimisation uses the hourly solar profiles and the flat-load heat demand on 16 representative days, obtained by k-means clustering of the 365 daily solar profiles and weighted by cluster size. [TODO: 8,760-hour validation sample and error.]

Selecting operating points and aggregation

To trace how cost changes with the share of heat served, the model is solved for every facility at availability floors from 10% to 100% in steps of 10%, for both the 5 km and 15 km land buffers. This gives each facility a curve of LCOH against availability. Where land limits solar capacity, the facility cannot reach the higher floors, and the Phase 2 solution gives the highest availability it can achieve. For each point, the heat served is the achieved availability multiplied by the facility's replaceable heat demand.

We then select one operating point per facility using its national gas heat benchmark. The *maximum heat* point is the highest availability at which the facility's LCOH is at or below its country's gas heat cost. The *maximum savings* point is the availability that maximises the saving relative to gas, the heat served multiplied by the difference between the gas heat cost and the LCOH. A facility whose LCOH is above the gas heat cost even at the lowest floor serves no heat below the benchmark.

For facility i, let α be the availability floor, $Q_i(\alpha)$ the heat served at that floor (the achieved availability multiplied by the replaceable heat demand), $\mathrm{LCOH}_i(\alpha)$ the cost per unit of heat served, and c_i^{gas} the gas heat cost of its country (Eq. 4). The two points are

$$\alpha_i^{\mathrm{heat}} = \max\left\{\alpha : \mathrm{LCOH}_i(\alpha) \le c_i^{\mathrm{gas}}\right\}, \tag{22}$$

$$\alpha_i^{\mathrm{sav}} = \arg\max_{\alpha}\ Q_i(\alpha)\left(c_i^{\mathrm{gas}} - \mathrm{LCOH}_i(\alpha)\right), \tag{23}$$

where the saving in Eq. 23 is evaluated only at floors with $\mathrm{LCOH}_i(\alpha) \le c_i^{\mathrm{gas}}$.

Heat served is summed across facilities and expressed as a share of total replaceable heat. Fleet and country LCOH are the means of facility LCOH weighted by heat served, which is equivalent to total annualised cost divided by total heat served.

Data availability

[TODO: data availability statement (note: BNEF data under licence)]

Code availability

[TODO: code availability statement (repository + DOI)]

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Author contributions

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Competing interests

[TODO: competing interests statement]

Figure legends

Figure 1: Modelled system configuration: co-located, off-grid solar PV paired with a heat pump/heat battery and battery storage supplying process heat to the facility. [TODO: add facility map panel?]

Figure 2: EU fleet-average LCOH vs. availability achieved at 5 km and 15 km land buffer zones, relative to the €93/MWh gas-heat and €171/MWh grid-electricity benchmarks.

Figure 3: Country-level LCOH vs. heat served at each country's gas-heat price, relative to €93/MWh: (a) 5 km, (b) 15 km

Figure 4: LCOH vs. availability for land-capped industries at 5 km and 15 km: (a) iron and steel, (b) steam cracking. Lines: gas €93/MWh, grid electricity €171/MWh. *[WISO26 Fig. 5:]* Facility-level heat availability achieved by land-constraint category: (a) 5 km buffer zone, (b) 15 km buffer zone

Table 1: Share of replaceable process heat served and levelised cost of heat (LCOH) at 5 km and 15 km buffer zones

	5 km	15 km
Served (highest served under gas prices)	58.8% (307 TWh)	84.4% (441 TWh)
LCOH (highest served under gas prices)	€55	€62
Served (max savings vs €93)	53.4% (278 TWh)	73.9% (377 TWh)
LCOH (max savings vs €93)	€41	€42

Table 2: Solar overbuild, capacity, land use, and LCOH for two high-temperature steel sites at 15 km buffer zone (90% availability)

	Solar/load	PV built	Land used	LCOH
Taranto	5.3×	3.0 GW	32%	€33
Linz	15.0×	8.5 GW	80%	€85

Extended Data

Extended Data Table 1: Specific energy consumption (SEC) and useful thermal efficiency (η) used in the production route, by sector. SEC is fuel input for process heat per tonne of product; feedstock is excluded. s_{fossil} is the EU-27 fossil share of fuel use from Eurostat nrg_d_indq_n (2024)[22]; country values replace it where Eurostat data are reliable.

Sector (product)	Route / type	SEC ($\mathrm{GJ\,t^{-1}}$)	η	s_{fossil}	Source
Cement (cement)	all	2.8	0.60	0.84	IEA/CSI GNR and IEA Cement Roadmap ($3.5\,\mathrm{GJ\,t^{-1}}$ clinker $\times$ 0.77 clinker ratio); η Madlool et al.[36]
Lime (lime)	all	4.5	0.65	0.84	EuLA; EU BREF Cement, Lime, MgO
Glass (glass)	all	6.5	0.45	1.00	Schmitz et al. 2011; Beerkens 2008
Pulp and paper (pulp, paper)	all	10.5	0.85	0.43	CEPI Key Statistics; US DOE steam boiler efficiency
Steam cracking (ethylene)	all	35.9	0.90	0.98	Hotmaps benchmark (total furnace duty); η Ren et al.[37]
Iron and steel (crude steel)	BF/BOF	4.8	0.70	1.00	Hotmaps (sinter + rolling); IEA Iron and Steel, worldsteel
	EAF	2.5	0.70	1.00	Hotmaps (fuel only, electricity excluded)
	DRI-EAF	12.0	0.70	1.00	Hotmaps (natural gas for DRI + EAF fuel)
Ammonia (NH_3)	chemicals	9.0	0.90	0.98	IEA Ammonia Technology Roadmap (reformer fuel only)
Soda ash (soda ash)	chemicals	10.0	0.85	0.98	EU BREF LVIC
Methanol (methanol)	chemicals	9.0	0.90	0.98	IEA/IRENA (reformer fuel only)
Alumina (alumina)	aluminium refinery	11.0	0.85	1.00	IAI; IEA
Aluminium (aluminium)	smelting	3.0	0.65	1.00	IAI (anode baking and casting; electricity excluded)

Extended Data Table 2: Parameters of the emissions route for sectors without production data (food, beverages and tobacco; textiles and leather).

Parameter	Value	Source
Natural gas emission factor	$56.1\,\mathrm{tCO_2\,TJ^{-1}}$	IPCC 2006 Guidelines, Vol. 2, Table 2.3[20]
Residual fuel oil emission factor	$77.4\,\mathrm{tCO_2\,TJ^{-1}}$	IPCC 2006 Guidelines, Vol. 2, Table 2.3[20]
Gas/oil fuel share	90%/10%	Eurostat nrg_d_indq[21]
Blended emission factor EF	$58.2\,\mathrm{tCO_2\,TJ^{-1}}$	$0.9 \times 56.1 + 0.1 \times 77.4$
Combustion share f_{comb}	1.0	No process CO_2 in food and textiles
Boiler efficiency η_{boiler}	0.85	Typical industrial steam boiler (US DOE)
Fossil-fuel share s_{fossil}	0.89 food; 0.95 textiles (EU-27)	Eurostat nrg_d_indq_n[22]

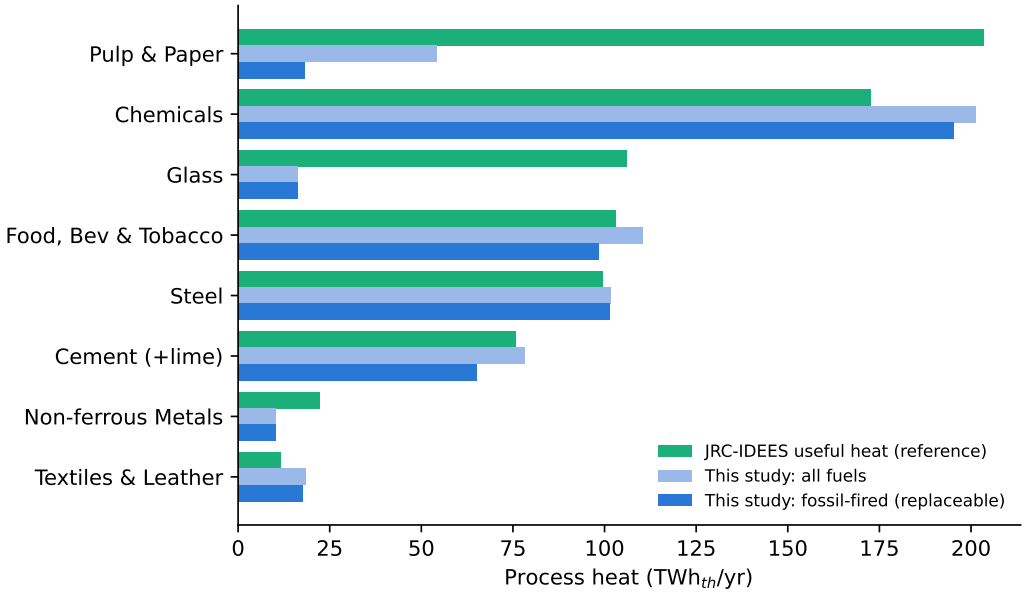
Extended Data Table 3: Key technology and system parameters. Costs are in real 2025 US dollars.

Parameter	Value	Source
PV CAPEX, FOM	by country (2030)	BNEF
Li-ion energy / power CAPEX	by country (2030)	BNEF
HP CAPEX, <100 °C	$\$831.73\,\mathrm{kW_{th}^{-1}}$ (2025 USD)	Rissman
HP CAPEX, 100–200 °C	$\$1{,}033.72\,\mathrm{kW_{th}^{-1}}$ (2025 USD)	Rissman
IHB CAPEX (in / out / storage)	$\$105.66$ / $\$316.99\,\mathrm{kW^{-1}}$; $\$5.28\,\mathrm{kWh^{-1}}$	Rissman & Gimon
COP (<100 / 100–200 °C)	3.7 / 2.2	Rissman
RTE (Li-ion / IHB)	0.90 / 0.95	
Max Li-ion duration $h_{\max}$	16 h	
Lifetime	25 yr	
PV power density	$50\,\mathrm{MW_{dc}\,km^{-2}}$	

Supplementary Information

Supplementary Note 1: Validation of heat estimates

To validate the heat estimates, we compared them against sector and country aggregates of useful heat from JRC-IDEES[23] (Supplementary Fig. 1). Estimates for steel, cement, food and chemicals are within ±15% of the reference, while the remaining sectors differ more. Three differences in scope explain most of the gap. First, JRC-IDEES covers heat from all fuels, whereas our replaceable heat counts only fossil-fired heat not used as feedstock, so biomass heat in cement and food appears in the reference but not in our estimate. The all-fuel bars in the figure remove this effect. Second, our estimate includes only facilities in the Climate TRACE dataset, which leaves pulp and paper and non-ferrous metals under-covered. Third, the JRC-IDEES ceramics and glass category also includes ceramics, brick and refractory firing, which we do not model, so our glass-only estimate is far lower. Textiles, which is estimated from emissions rather than production, is the least certain sector. These heat estimates carry further limitations. They use 2024 production and emissions, whereas costs are set for 2030, and demand is modelled as a flat load throughout the year, so any seasonal or daily variation in process heat is not captured. This uncertainty in heat demand is a known limitation, but because we compute the levelised cost of heat (LCOH) per unit of heat delivered, the results depend less on the absolute size of the heat estimate than a total-demand figure would.



Supplementary Fig. 1: Validation of bottom-up process heat against JRC-IDEES useful heat for the EU-27, by sector. Bars show the JRC-IDEES reference (all fuels), this study's total heat from all fuels, and this study's fossil-fired (replaceable) heat.

Supplementary Table 1: Symbols used in the optimization model.

Symbol	**Meaning**
$k,\ K$	Representative day and number of representative days (16)
$t,\ H$	Hour of the day and hours per day ($H = 24$, with $t - 1 \equiv H$ when $t = 1$)
$\mathcal{B}^{\mathrm{HP}},\ \mathcal{B}^{\mathrm{IHB}}$	Bands served by heat pump (COP > 1; <100 and 100–200 °C) or by IHB (COP ≤ 1; ≥200 °C)
$g_{k,t}$	Solar capacity factor in hour t of representative day k (MW per MW installed)
L_b	Flat heat load of band b ($\mathrm{MW_{th}}$)
σ_t	Diurnal load shape, mean 1 (all 1 if flat)
$y_{b,k,t}$	IHB heat delivered to band b
$p^{\mathrm{ch}}_{j,k,t},\ p^{\mathrm{dis}}_{j,k,t},\ s_{j,k,t}$	Charge, discharge and state of charge of storage j
η_j	One-way efficiency of storage j, $\sqrt{\mathrm{RTE}_j}$
$h_{\max}$	Maximum storage duration, 16 h by default
AF	Availability factor, which sets the share of heat that must be served
$P^{\max}_{\mathrm{PV}}$	Optional solar cap from land limits (no cap if unset)
H	Number of hours in representative days
w_k	Weight of representative day k (number of days it represents)
$x^{\mathrm{dir}}_{b,k,t},\ x^{\mathrm{batt}}_{b,k,t}$	Electricity to the heat pump of band b directly from solar or from the Li-ion battery
$u_{b,k,t}$	Unserved heat in band b
COP_b	Heat pump coefficient of performance for band b
$P^{\mathrm{HP}}_b,\ P^{\mathrm{ch}}_j,\ P^{\mathrm{dis}}_j,\ E_j$	Heat pump capacity, storage charge and discharge power, and storage energy capacity
$c_{\mathrm{PV}},\ c_{\mathrm{Li},E},\ c_{\mathrm{Li},P},\ c_{\mathrm{IHB},\cdot},\ c^{\mathrm{HP}}_b$	Annualised unit costs of each technology
RTE_j	Round-trip efficiency of storage j (0.90 Li-ion, 0.95 IHB)
x	Electricity